# Markscheme

Mathematics: analysis and approaches
Practice question bank

---

**5. (a)**

Given $a = 7 \times 10^{-4}$ and $b = 2 \times 10^{-5}$.

Calculate $C = \sqrt[3]{a - b}$, giving your answer in the form $k \times 10^{n}$ correct to 3 significant figures.

match the powers of 10 before subtracting: $a = 7 \times 10^{-4} = 70 \times 10^{-5}$ *(M1)*

$$a - b = 68 \times 10^{-5} = 6.8 \times 10^{-4}$$ **A1**

rewrite with a power of 10 divisible by 3: $6.8 \times 10^{-4} = 680 \times 10^{-6}$ *(M1)*

attempt to take the cube root *(M1)*

$$C = \sqrt[3]{680 \times 10^{-6}} = \sqrt[3]{680} \times 10^{-2}$$ **A1**

$$C \approx 8.79 \times 10^{-2}$$ **A1**

**Note:** Do not accept a final answer given in calculator notation, e.g. $8.79E{-}2$.

**[6 marks]**

**(b)**

Hence write $C^3$ in the form $k \times 10^{n}$, and verify it approximately equals $a - b$.

$$C^3 = (8.79 \times 10^{-2})^3 \approx 6.80 \times 10^{-4}$$ *(M1)*

this matches $a - b$ from part (a), as required **A1**

**[2 marks]**